D Y PATIL DEEMED TO BE UNIVERSITY
RAMRAO ADIK INSTITUTE OF TECHNOLOGY
NAVI MUMBAI

Semester – V **Academic Year 2025-26 (ODD)**

Assignment-II

Course Code: 2413ITU3T1 /2413CEU3T1 /2413CBU3T1 /2413ADU3T1 /2413CAU3T1 / 2413CCU3T1/ 2413ECU3T1

Course Name: Embedded systems Branch: CE/CSBS/AIML/AIDS/CS/IT/ECE

Maximum Marks: 10 Submission Date: 19/09/2025

Note for Students:
All the Questions are compulsory and carry equal marks .

Question	Solve all questions	Max. Marks	CO	BT
Q1 a	Write a program to find whether the given number is EVEN or ODD. Move "00H" to R0 register if the number is EVEN and Move "0FFH" to R0 register if the number is ODD.	3	CO3	BT4
Q1 b	Explain AJMP,LJMP, SJMP instructions in detail as well as the difference between them.	2	CO3	BT2
Q2 a	Write a program to generate a time delay of 2 seconds using Timer-0 of 8051 microcontroller.	3	CO4	BT4
Q2 b	Explain SWAP, XCH, XCHD instructions in 8051 microcontroller.	2	CO4	BT2

Course Outcomes (CO) -Learner will be able to:

CO1: Understand number systems, logic gates, and design basic combinational and sequential circuits.
CO2: Explain embedded systems, their classifications, applications, and key architectural concepts.
CO3: Analyze 8051 architecture and apply assembly programming concepts.
CO4: Develop programs using I/O ports, timers, serial communication, and interrupts.
CO5: Design and implement 8051-based peripheral interfacing and real-world embedded applications.
CO6: Explain ARM architecture, register operations, pipelining, and evaluate Cortex processor architecture.

Bloom's Taxonomy:

BT1- Remembering, BT2- Understanding, BT3- Applying, BT4- Analyzing, BT5- Evaluating, BT6- Creating

Subject In charge DQA HOD

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